Jared Siegel | Curriculum Vitae

University of Chicago, Department of Astronomy & Astrophysics — Chicago, IL 60637

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Education

The University of Chicago | Chicago, IL — Expected June 2022

Bachelor of Arts in Physics
Bachelor of Science in Astrophysics

Highlands Ranch High School | Highlands Ranch, CO — 2014 to 2018

Diploma
GPA: 4.40/4.00 Weighted | GPA: 4.00/4.00 Unweighted | 840 Community Service Hours

Research Experience

Investigating Detection Biases in the X-ray Binary Population — Jun. 2020 to Present
Northwestern University NSF CIERA REU | Supervised by Vicky Kalogera and Christopher Berry

- Executed population synthesis studies of binary formation and evolution
- Wrote and operated post-processing scripts for inferring system properties and detection probabilities

Exploring Tidal-Dissipation and Critical-Angles in Resonance Chains — Nov. 2019 to Present
University of Chicago Dept. of A. & A. | Supervised by Daniel Fabrycky

- Investigated the formation and evolution of exoplanet systems through N-body simulations
- Wrote and executed `python` scripts for studying exoplanet dynamics in the simulation data

Analyzing Supernova Remnant X-ray Emission — Jan. 2019 to Sept. 2020
University of Chicago Dept. of A. & A. | Supervised by Vikram Dwarkadas and Kari Frank

- Implemented new methods for chemical abundance calculations within supernova remnants
- Wrote and executed `python` code to analyze XMM Newton X-ray observations of supernova remnants
- Synthesized results and drafted papers for publication

Publications

Refereed Publications

3. "Resonant Chains of Exoplanets: Libration Centers for Laplace Angles'

   Siegel J, Fabrycky D, submitted to AJ.

2. "Analysis of XMM-Newton Observations of Supernova Remnant W49B and Clues to the Progenitor"

   Siegel J, Dwarkadas VV, Frank K, Burrows DN. 2020, accepted to ApJ.
   https://ui.adsabs.harvard.edu/abs/2020arXiv201004765S/abstract

1. "Smoothed particle inference analysis and abundance calculations of DEM L71, and comparison to SN explosion models"

   Siegel J, Dwarkadas VV, Frank K, Burrows DN, Panfichi A. Astron. Nachr. 2020; 1–7.
   https://ui.adsabs.harvard.edu/abs/2020arXiv200110129S/abstract

Articles

1. "Elemental Abundances in Supernova Remnant W49B as Clues to Its Progenitor"

   Siegel J, Dwarkadas VV, Frank K, Burrows DN. 2020, RNAAS
   https://iopscience.iop.org/article/10.3847/2515-5172/abac07

Abstracts

2. "SPI Analysis and Abundance Calculations of W49B"

   Siegel, J., Dwarkadas, V., Frank, K., & Burrows, D. 2020,in American Astronomical Society Meeting Abstracts,Vol. 236, American Astronomical Society Meeting Abstracts 236, 134.032
   https://ui.adsabs.harvard.edu/abs/2020AAS...23613403S/abstract

1. "SPI Analysis and Abundance Calculations of DEM L71 and W49B, and Comparison to SN explosion Models"

   Frank, K. A., **Siegel, J.**, Dwarkadas, V., Burrows, D. N., & Panfichi, A. 2020, in American Astronomical Society Meeting Abstracts, American Astronomical Society Meeting Abstracts, 377.02
   https://ui.adsabs.harvard.edu/abs/2020AAS...23537702F

Presentations

4. "Smoothed Particle Inference Studies of Supernova Remnant DEM L71"

   University of Chicago Research Symposium Proceedings | Oral Presentation | June 2020
   https://openjournals.lib.uchicago.edu/index.php/ugradresearchsymposium/article/view/53/26

3. "Smoothed Particle Inference Studies of Supernova Remnant Abundances"

   Midstates Consortium for Math and Science | Poster Presentation | November 2019
   UCISTEM Undergraduate Research Symposium | Poster Presentation | October 2019

2. Keynote Panelist

   UCISTEM Undergraduate Research Symposium | Keynote Panel and Workshop | October 2019

1. "Smoothed Particle Inference Studies of Supernova Remnant Abundances"

   UCISTEM Undergraduate Research Symposium | Poster Presentation | October 2019

Awards and Grants

College Research Fellows (Hoeft) Award | University of Chicago — November 2020 to June 2021

Chambliss Astronomy Student Award | 236th Meeting of the American Astronomical Society — June 2020

Micro-Metcalf Grant | University of Chicago — Spring 2020

Summer Action Grant | University of Chicago — June 2019

University Scholar Award | University of Chicago — 2018 to Present

Skills

- Advanced knowledge of `python` programming
- Moderate knowledge of `C`, `HTML`, and `Bash` programming
- Extensive experience with cluster computing

Additional Experience

Board Member | Society of Physics Students — March 2020 to Present

- Planned and executed remote lecture series fundraiser for University of Chicago Emergency Fund
- Contributed to organization wide budgeting and event planning

Student Coordinator | Taking the Next Step Conference — Sept. 2019 to Present

- Led alumni outreach for the Research Round-tables and Panel
- Wrote panel questions and led panel event

Writer | The University of Chicago Chapter of Triple Helix — April 2019 to April 2020

- Unraveling Supernova Remnant DEM L71
- Interstellar Visitors
- The Birth of Extra-Galactic Astronomy: How Seventy-Four Light Bulbs Forever Changed Our View of the Universe

Student | Tutorials on Mechanical Design for Scientific Apparatus — Summer 2019

- Completed a training sequence with the Physics Department covering CAD design, 3-D printing, and machining

Student | Experimental Particle Physics Reading Seminar — Winter 2019

- Engaged in a weekly Reading Seminar covering ongoing experiments in particle physics and the detection of dark matter
- Lead a session covering the role of gravitational lensing in the study of dark matter

References

Daniel Fabrycky | Associate Professor | University of Chicago
(773) 702-9562
fabrycky@uchicago.edu

Vikram Dwarkadas | Research Professor | University of Chicago
(773) 834-4668
vikram@uchicago.edu

Kari Frank | Director of Operations of CIERA, Research Assistant Professor | Northwestern University
(847) 467-3178
kari.frank@northwestern.edu